CHAPTER

9

MATHEMATICS

HYPERBOLIC FUNCTIONS

SYNOPSIS AND FORMULAE

Hyperbolic functions are similar to trigonometric functions. Some of the hyperbolic functions are,

1. Hyperbolic sine of x i.e. $\boxed{\sin hx = \frac{e^x - e^{-x}}{2}}$

   It is an odd function i.e., $f(-x) = f(-x)$

2. Hyperbolic cosine of x i.e., $\boxed{\cosh x = \frac{e^x + e^{-x}}{2}}$

   It is an even function i.e., $f(-x) = f(x)$

3. Hyperbolic tangent of x i.e., $\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{\sinh x}{\cosh x}$

   It is an odd function.

4. Hyperbolic secant of x i.e., $\operatorname{sech} x = \frac{2}{e^x + e^{-x}} = \frac{1}{\cosh x}$

   It is an even function.

5. Hyperbolic cotangent of x i.e., $\operatorname{cosec} hx = \frac{2}{e^x - e^{-x}} = \frac{1}{\sinh x}$

   It is an odd function.

6. Hyperbolic contagent of x i.e., $\coth x = \frac{e^x + e^{-x}}{e^x - e^{-x}} = \frac{1}{\tanh x} = \frac{\cosh x}{\sinh x}$

   It is an odd function.

Hyperbolic functions:
i. Sinhx
ii. coshx

Note

1. $\sin h(0) = 0$
2. $\cos h(0) = 1$
3. $\tan h(0) = 0.$

Hyperbolic Function Identities

1. $\cos h^2 x - \sinh^2 x = 1$
2. $\sec^2 hx + \tanh^2 x = 1$
3. $\cot h^2 x - \operatorname{cosech}^2 x = 1$

- The domain and range of hyperbolic functions are illustrated in table below,

Domain and Range of Hyperbolic Functions

S.No.	Function $y = f(x)$	Domain (x)	Range(y)
(i)	y = sinh(x)	R	R
(ii)	y = cosh(x)	R	[1, ∞]
(iii)	y = tanh(x)	R	(–1, 1)
(iv)	y = coth(x)	R – {0}	(–∞, –1) ∪ (1, ∞)
(v)	y = sech(x)	R	(0, 1)
(vi)	y = cosech(x)	R – {0}	R – {0}

+ Formulae related to hyperbolic functions.

1. $\sin h\,(x \pm y) = \sin hx \cosh y \pm \cosh x \sin hy$
2. $\cos h\,(x \pm y) = \cos hx \cosh y \pm \sinh x \sin hy$
3. $\tan\,(x \pm y) = \dfrac{\tanh x \pm \tanh y}{1 \pm \tanh x \tanh y}$
4. $\sin h2x = 2\sin hx \cosh x = \dfrac{2\tanh x}{1-\tanh^2 x}$
5. $\cos h2x = \cos h^2x + \sin h^2x = 1 + 2\sinh^2 x = 2\cosh^2 x - 1$
6. $\tan h2x = \dfrac{2\tanh x}{1+\tanh^2 x}$
7. $\sin h3x = 3\sin hx + 4\sin h^3x$
8. $\cos h3x = 4\cos h^3x - 3\cos hx$
9. $\tan h3x = \dfrac{3\tanh x + \tanh^3 x}{1+3\tanh^2 x}$

+ Inverse hyperbolic functions are defined as the inverse of hyperbolic functions. Some of the inv hyperbolic functions are,

1. Inverse hyperbolic sine of x.

   i.e., $\sin h^{-1} x = \log_e\left(x+\sqrt{x^2+1}\right)$

2. Inverse hyperbolic cosine of x.

   i.e., $\cos h^{-1} x = \log_e\left(x+\sqrt{x^2-1}\right)$

3. Inverse hyperbolic tangent of x.

   i.e., $\tanh^{-1} x = \dfrac{1}{2}\log_e\left(\dfrac{1+x}{1-x}\right)$

4. Inverse hyperbolic secant of x.

   i.e., $\mathrm{sech}^{-1} x = \log_e\left(\dfrac{1+\sqrt{1-x^2}}{x}\right)$

5. Inverse hyperbolic cosecant of x.

   i.e., $\mathrm{cosech}^{-1} x = \log_e\left(\dfrac{1+\sqrt{1+x^2}}{x}\right)$

6. Inverse hyperbolic cotangent of x.

   i.e., $\cot h^{-1} x = \dfrac{1}{2}\log_e\left(\dfrac{x+1}{x-1}\right)$

+ The domain and range of inverse hyperbolic functions are illustrated in table below,

Domain and Range of Inverse Hyperbolic Functions

S.No.	Inverse Hyperbolic Function, y = f(x)	Domain (x)	Range(y)
(i)	$y = \sinh^{-1}(x)$	R	R
(ii)	$y = \cosh^{-1}(x)$	$[1, \infty)$	$[0, \infty)$
(iii)	$y = \tanh^{-1}(x)$	$(-1, 1)$	R
(iv)	$y = \coth^{-1}(x)$	$R - [-1, 1]$	$R - \{0\}$
(v)	$y = \mathrm{sech}^{-1}(x)$	$(0, 1]$	$[0, \infty)$
(vi)	$y = \mathrm{cosech}^{-1}(x)$	$R - \{0\}$	$R - \{0\}$